
Chapter 15. 4690 Directory Services

This chapter explains 4690 OS support for directory services. Directory Services has been added to 4690 OS controllers in order to support enterprise managed user IDs. The 4690 logon process has been enhanced so that it can access the LDAP directory for user ID information, including password and authorization attribute data. In simple terms, a directory is just a database. Directories are usually optimized for reading data. 4690 OS uses Lightweight Directory Access Protocol (LDAP) for accessing the directory. The 4690 OS implementation is provided by OpenLDAP, an open source LDAP implementation. For more information about OpenLDAP and directory services, go to www.openldap.org.

Directory Services requirements

Directory Services runs only on 4690 OS controllers.

Configuring Directory Services

Directory Services is configured in System Configuration. There is a panel labeled "Directory Services" in the System Security section. Select the check box to enable Directory Services, then activate System Configuration. Directory Services will be started the next time you restart 4690 OS provided you have a valid `ADX_IDT1:ADXLDIFF.DAT` file, as described below.

Deselecting the check box and activating System Configuration will disable Directory Services, and Directory Services will continue to operate until 4690 OS is restarted.

Warning - when the 4690 OS is restarted, all data stored in the directory is deleted.

Initializing the directory

After enabling Directory Services in System Configuration, additional configuration data must be provided. Directory Services looks for that data in `ADX_IDT1:ADXLDIFF.DAT`. Note that the size of `ADXLDIFF.DAT` is limited to 65,536 bytes. A sample file, `ADX_IDT1:ADXLDIFF.SMP` is provided. The sample file contains extensive commenting to help you create your own `ADXLDIFF.DAT` file. Some of those comments will be reproduced below.

When 4690 OS initializes LDAP, it saves a copy of `ADX_IDT1:ADXLDIFF.DAT` internally. On each subsequent reboot, the internal version number is compared to the version number in `ADX_IDT1`. If a version change is detected, the existing LDAP directory is erased, and LDAP is re-initialized using the data from `ADX_IDT1:ADXLDIFF.DAT`. The new version becomes the one saved internally.



Note: If LDAP has been initialized, and `ADX_IDT1:ADXLDIFF.DAT` is not found, by default there is no version change. For security reasons, we suggest that you remove user definitions from `ADX_IDT1:ADXLDIFF.DAT` after LDAP has been initialized. Do not remove the file, as it may be needed during a LAN Disk Rebuild to setup a new controller.

Verifying installation

During initialization, the LDAP utility *slapadd* is used to populate the directory with entries from `ADXLDIFF.DAT`. Output from *slapadd* is saved in `F:\adxetc/logs/slapadd.log`. The results from adding each entry are logged. If any of the add commands fail, initialization fails and the LDAP database remains empty. Correct the entry and restart the controller to rerun the installation process.

ADXLDIFF.DAT

ADXLDIFF.DAT has two sections. The first section is the header. The header contains configuration attributes that help 4690 OS setup the LDAP directory. The second section contains initial entries for your directory, specified in LDAP Interchange format. The entries are grouped by object class. Comments are provided in ADXLDIFF.DAT for each class, and for each entry. Entries required by 4690 OS are noted.

The ADXLDIFF.DAT header

As mentioned above, the header contains configuration attributes used by 4690 OS to set up the LDAP directory. The sections below detail the required and optional configuration attributes.

Required attributes

version: 001

This is a character string indicating the current version of the ADXLDIFF.DAT file. Any change in this string signals a change of version. If a version change is detected, the existing LDAP directory is erased, and LDAP is re-initialized using the data from ADX_IDT1:ADXLDIFF.DAT. The new version becomes the one saved internally.

root: dc=myCorp,dc=com

This is the top entry in the LDAP directory, also called the base or the suffix. The distinguished name of every directory entry contains the root. Here are two examples:

```
a.    dc=toshiba,dc=com
b.    l=raleigh,o=toshiba,c=us
```

admin: cn=administrators,ou=groups,dc=myCorp,dc=com

This is the distinguished name of the Administrator group. The Administrator group is a special groupOfNames object in the 4690 OpenLDAP implementation. You **must** be a member of this group to have read/write access to the 4690 LDAP database. The Administrator group must be defined in this file, using the same distinguished name. See ["The Initial Administrator" on page 312](#) for more details.

security: level=simpletls,strength=112

This sets the security level for communications between the LDAP server and clients. The level can be set to one of ~~three~~ values.

1. level=none indicates no communication security will be used. This is not recommended.
2. Set level=simpletls, strength=xxx to use a simple TLS (Transport Layer Security) connection. xxx indicates the encryption key length. Set it to an integer between 40 and 256. 112 is equivalent to 3DES. When level is set to simpletls, 4690 Directory Services will generate self-signed certificates for the LDAP server. LDAP clients attempting to connect to the 4690 LDAP server must have a certificate so an encryption method can be negotiated, but no certificate checking is done.
3. Set level=tls, strength=xxx to use a TLS connection. xxx indicates the encryption key length. Set it to an integer between 40 and 256. 112 is equivalent to 3DES. When level is set to tls, the user is expected to provide the certificates used during TLS authentication:

- `f:/adxetc/ext/adxslapd/cacert.pem` - filename of the Certificate Authority certificate. If the file cannot be found, initialization fails.
- `f:/adxetc/ext/adxslapd/xxcert.pem` - filename of the LDAP server certificate, where xx is the controller ID. If the file cannot be found, initialization fails.
- `f:/adxetc/ext/adxslapd/xxkey.pem` - filename that contains the private key matching the LDAP server certificate, where xx is the controller ID. If the file cannot be found, initialization fails.

defaultpp: cn=default,ou=policies,dc=myCorp,dc=com

This is the distinguished name of the default password policy. You may specify a password policy to use in each user definition. If a user does not have a defined password policy specified in the `pwdPolicySubentry` attribute, LDAP will use the default password policy. If you do not want a default password policy, enter `defaultpp: none`. See ["Password Policy Settings" on page 314](#) for more details.

clientcertificates: no

This indicates whether the user is providing client certificates to the 4690 OS. Client certificates are typically configured in the `servers ldap.conf` file. 4690 Directory Services code that accesses the LDAP database could be required to authenticate using the client certificates, depending on your configuration settings. The `clientcertificates` attribute is ignored unless the security level is set to `tls`. The attribute has three possible settings:

1. `yes` - the user will provide client certificates. If the following file names are not found, initialization will fail:
 - `f:/adxetc/ext/adxslapd/xxclcert.pem` - Filename of the client certificate, where xx is the controller ID.
 - `f:/adxetc/ext/adxslapd/xxclkey.pem` - Filename that contains the private key matching the client certificate, where xx is the controller ID.
2. `no` - client certificates will not be used.
3. `server` - the server certificates configured above will also be used as client certificates.

verifyserver: never

Must be set to `never`, `allow`, `try`, or `demand`, as described in the *OpenLDAP Administrator's Guide*. Set this to `demand` for the strictest security requirements.

verifyclient: never

Must be set to `never`, `allow`, `try`, or `demand`, as described in the *OpenLDAP Administrator's Guide*. Set this to `demand` for the strictest security requirements. Note that setting `clientcertificates` to `no`, then setting `verifyclient` to `demand` is an invalid configuration and will cause initialization to fail.

Optional attributes

The following attributes direct 4690 client code to connect to the configured LDAP server instead of the local server. All of these attributes must be specified, or none of them. In the sample file, none are specified; they are commented out.

uri: ldaps://server.myCorp.com:636

The name of the server to connect to.

binddn: cn=Default Administrator,ou=people,dc=myCorp,dc=com

The Distinguished Name used to bind to the server. This user must have read access to the 4690 user information on the server.

credentials: epw:xyyzz

The password used to bind to the server. The prefix `epw:` must be present. `xyyzz` is the encrypted password. It must be generated by the 4690 OS utility `ADXLDEPW.286`.

See [“Configuring an External Directory” on page 316](#) for more details.

Optional attributes can be used to add configuration options that apply to the LDAP client. In this example, the line `timeout 5` is added to the client configuration file `.ldaprc`.

clientcfg.timeout: 5

For more details, see [“Client configuration options” on page 317](#).

The line `@@End of configuration header` signals the end of the configuration section. Do not remove this line from your `ADXLDIFF.DAT` file.

ADXLDIFF.DAT Directory Entries

After the header, you define the initial entries for your directory using LDIF, LDAP Data Interchange Format. LDIF is simply a way to specify a directory entry in a plain text format.

We suggest you limit the directory entries defined in `ADXLDIFF.DAT` to those entries required to get the directory up and running. The sample file includes the following types of entries:

- Root entries (required)
- An administrator group (required)
- An initial administrator (required)
- 4690 User Authorization roles and groups (recommended)
- Password Policy settings (optional)



Note: The contents of `ADXLDIFF.SMP` is subject to change. The most current version, with detailed documentation, can be found in the `ADX_IDT1` directory.

Root Entries

LDAP directory entries are organized in a tree-like structure. You will need to define the root entries used by the other entries in the file. For example, the [Figure 17](#) sample file defines the following tree structure: